Discrete Mathematical Structures

Course Code: **NMA216** • BS-MS (Mathematics & Data Science)
Harcourt Butler Technical University, Kanpur

Complete Exam Guide — Theory + Solved Papers + 10 Probable Questions

Made for your Mid-Sem exam. Every idea is explained in simple, plain language first, then used to solve the actual questions. Read top to bottom like a mini-textbook. You've got this! 💪

How to read this guide

- **Yellow box** = the idea in super-simple words.
- **Blue box** = the exact definition / formula to write in the exam.
- **Green box** = the final answer.

Notation: $\overline{x}$ (a bar on top) means **NOT x** (complement). $a \mid b$ means "a divides b". $\mathbb{Z}$ = integers, $\mathbb{Q}$ = rationals, $\mathbb{R}$ = real numbers, $\mathbb{R}^*$ = non-zero reals.

PART A — Theory (everything you need, in easy language)

A1. What is a Group? (CO3)

In easy words: A group is just a **set of things** plus **one rule to combine two of them** (like + or ×), where the rule behaves nicely — you never fall out of the set, you can re-bracket freely, there's a "do-nothing" element, and everything can be undone.

Definition: A set **G** with an operation ***** is a **group** if it satisfies 4 axioms:

1. **Closure:** for all a, b in G, a * b is also in G.
2. **Associativity:** (a * b) * c = a * (b * c).
3. **Identity:** there is an element e in G with a * e = e * a = a for all a.
4. **Inverse:** for each a there is a^{-1} in G with $a * a^{-1} = a^{-1} * a = e$.

If also **a * b = b * a** for all a, b, the group is **abelian (commutative)**.

How to CHECK if something is a group (exam recipe): test the 4 axioms in order — Closure → Associativity → Identity → Inverse. If even one fails, it is **not** a group. Mention commutativity only as a bonus.

Order of a group = number of elements in it. **Order of an element a** = smallest positive n with $a^n = e$.

A2. Subgroups

In easy words: A subgroup is a smaller group living inside a bigger group, using the same operation.

Definition: H is a subgroup of G (written $H \leq G$) if $H \subseteq G$ and H is itself a group under G's operation.
Subgroup Test (quick): H (non-empty) is a subgroup if for all a, b in H, $\mathbf{a * b^{-1} \in H}$.

A3. Cosets & Normal Subgroups (CO3)

In easy words — coset: Take a subgroup H and "slide" it by an element g. The slid copy $gH = \{ g*h : h \in H \}$ is a **left coset**; Hg is a **right coset**. Cosets neatly cut the whole group into equal-sized pieces.

In easy words — normal: A subgroup is **normal** if sliding it from the left gives the same set as sliding from the right (gH = Hg). This is exactly the condition that lets us build a "quotient group" G/H.

Definition: A subgroup N of G is **normal** (written $N \triangleleft G$) if for every $g \in G$,
$gN = Ng$, equivalently $\mathbf{g\ n\ g^{-1} \in N}$ for all $n \in N$.

Golden fact for the exam: In an **abelian group, EVERY subgroup is normal** (because $g\ n\ g^{-1} = n\ g\ g^{-1} = n$). This single line solves the "even integers" question.

Example of a normal subgroup: $3\mathbb{Z} = \{\ldots,-6,-3,0,3,6,\ldots\}$ is a normal subgroup of $(\mathbb{Z}, +)$ because $(\mathbb{Z},+)$ is abelian. Another classic: the centre of any group, or A_n inside S_n.

A4. Group Homomorphism & Isomorphism (CO3)

In easy words — homomorphism: a map between two groups that **respects the operation** — "combine then map" = "map then combine".

Homomorphism: $f: G \to G'$ with $\mathbf{f(a * b) = f(a) \cdot f(b)}$ for all $a, b \in G$ (left side uses G's operation, right uses G′'s).
Kernel: $\ker f = \{ a \in G : f(a) = e' \}$. The kernel is always a **normal subgroup**. f is **one-to-one (injective)** $\Leftrightarrow$ **ker f = {e}**.

In easy words — isomorphism: a homomorphism that is also a **perfect pairing** (one-to-one AND onto). It means the two groups are "the same group wearing different clothes" — identical structure.

Isomorphism: a homomorphism f that is **bijective** (injective + surjective). Then $G \cong G'$.
To **prove** an isomorphism: (1) show it's a homomorphism, (2) show one-to-one, (3) show onto.
To **disprove** isomorphism: show it fails one-to-one (kernel bigger than {e}) OR fails onto.

A5. Relations, Partial Orders & POSETs (CO4)

In easy words: A "partial order" is a way of saying "this is ≤ that" for some pairs — like divisibility or subset — where not every two things need to be comparable (hence partial).

A relation R on set S is a partial order if it is:

1. **Reflexive:** a R a for all a.
2. **Antisymmetric:** if a R b and b R a then a = b.
3. **Transitive:** if a R b and b R c then a R c.

A set with a partial order is a **POSET** (partially ordered set), written (S, ≤).

A6. Hasse Diagram & Special Elements (CO4)

In easy words: A Hasse diagram is a clean picture of a POSET. Draw smaller things lower, bigger things higher, and connect only **direct** steps (no arrows, no self-loops, no transitive shortcuts).

Special elements:

- **Maximal:** nothing is strictly above it (top of a branch). There can be several.
- **Minimal:** nothing is strictly below it (bottom of a branch). There can be several.
- **Greatest (maximum):** ONE element that is ≥ every other element. Exists only if it's unique.
- **Least (minimum):** ONE element that is ≤ every other element.

Trap: "maximal" ≠ "greatest". Two incomparable tops ⇒ two maximal elements but **no** greatest element.

A7. Lattices (quick, CO4)

In easy words: A lattice is a POSET where every pair of elements has a single "nearest common ceiling" (LUB / join, written ∨) and a single "nearest common floor" (GLB / meet, written ∧).

For divisibility posets: join = LCM, meet = GCD. The divisors of n under "|" always form a lattice.

A8. Boolean Algebra & Functions (CO4)

In easy words: Boolean algebra works with only 0 and 1. "+" means OR, "·" (or just writing letters together) means AND, and a bar $\bar{x}$ means NOT.

Key laws: $x + 0 = x$, $x \cdot 1 = x$, $x + \bar{x} = 1$, $x \cdot \bar{x} = 0$, $x + x = x$, $x \cdot x = x$ (idempotent).
De Morgan: $\overline{x + y} = \bar{x} \cdot \bar{y}$ and $\overline{x \cdot y} = \bar{x} + \bar{y}$.

Minterm: an AND of every variable, each either plain or barred (e.g. $x\bar{y}z$). Each minterm is 1 for exactly one input row.
SOP (Sum of Products) / minterm expansion: OR together the minterms for every row where f = 1.
POS (Product of Sums) / maxterm expansion: AND together the maxterms for every row where f = 0.

A9. Quine–McCluskey Method (CO4)

In easy words: It's a neat, tabular way to simplify a Boolean function (a "by-hand, step-by-step K-map"). You list the minterms in binary, group them, and repeatedly combine terms that differ in exactly one bit — that one differing bit gets erased (becomes a dash "–"). Terms that can't combine any further are **prime implicants**. Finally pick the fewest prime implicants that cover all minterms.

Steps:

1. Write each minterm in binary; group by the **number of 1s**.
2. Compare terms in adjacent groups. If they differ in exactly ONE bit, combine them and put "–" in that position. Tick (✓) both terms used.
3. Repeat combining the new terms (same rule, and the dash positions must match).
4. Any term never ticked is a **Prime Implicant (PI)**.
5. Build a **PI chart** (PIs vs minterms). A minterm covered by only one PI makes that PI **essential**. Choose all essential PIs, then cover any leftover minterms with the fewest remaining PIs.

PART B — Solved Paper: MID TERM-II (2024-25)

Q1a · 2M · CO3 Define normal subgroup with an example. Check whether the set of even integers is a normal subgroup of the set of all integers under usual addition.

Solution

Definition: A subgroup N of a group G is **normal** if gN = Ng for every g ∈ G; equivalently g n g^{-1} ∈ N for all g ∈ G, n ∈ N. **Example:** 3ℤ = {…,−3,0,3,6,…} is a normal subgroup of (ℤ, +).

Now the even integers 2ℤ = {…,−4,−2,0,2,4,…} inside (ℤ, +):

Step 1 — is 2ℤ a subgroup? Even + even = even (closure ✓); 0 is even (identity ✓); the negative of an even number is even (inverse ✓); addition is associative ✓. So 2ℤ is a subgroup.

Step 2 — is it normal? (ℤ, +) is **abelian**. In an abelian group the "conjugate" is g + n + (−g) = n ∈ 2ℤ, so g + 2ℤ = 2ℤ + g for every g. Condition satisfied.

Answer: Yes — the even integers 2ℤ form a **normal subgroup** of (ℤ, +). (In fact every subgroup of an abelian group is normal.)

Q1b · 3M · CO4 Show that the set of all divisors of 30 under usual divisibility forms a POSET. Draw the Hasse diagram and find the maximal and minimal elements.

Solution

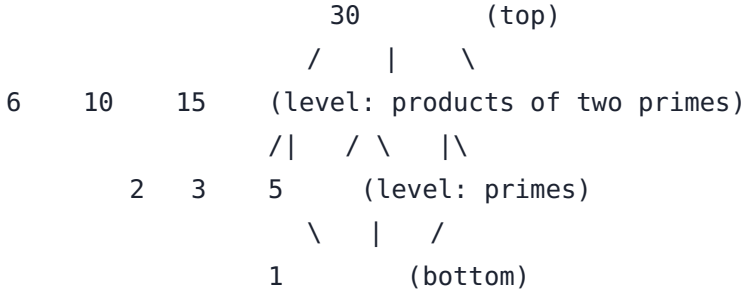
Divisors of 30: **D_{30} = {1, 2, 3, 5, 6, 10, 15, 30}**. Relation: a ≤ b means **a | b** (a divides b).

Is it a POSET?

- **Reflexive:** every number divides itself (a | a). ✓
- **Antisymmetric:** if a | b and b | a with positive integers, then a = b. ✓
- **Transitive:** if a | b and b | c then a | c. ✓

All three hold, so (D_{30}, |) is a **partial order** ⇒ it is a POSET.

Hasse diagram (lower divides upper; only direct cover edges drawn):



Cover edges: 1–2, 1–3, 1–5; 2–6, 2–10, 3–6, 3–15, 5–10, 5–15; 6–30, 10–30, 15–30.

Answer: It is a POSET. **Minimal element = 1** (also the least element). **Maximal element = 30** (also the greatest element).

Q2 · 5M · CO3 Define group isomorphism. Let $\mathbb{R}^*$ be the group of non-zero reals under multiplication and GL(2, $\mathbb{R}$) the group of 2×2 real matrices with non-zero determinant. Show that $A \mapsto \det A$ is a homomorphism from GL(2, $\mathbb{R}$) to $\mathbb{R}^*$ but **not** an isomorphism.

Solution

Definition (isomorphism): A map f: G → G′ is an isomorphism if it is a homomorphism — f(a*b) = f(a)·f(b) — and is **bijective** (one-to-one and onto). Then G ≅ G′.

Let φ: GL(2, $\mathbb{R}$) → $\mathbb{R}^*$ be φ(A) = det A. (Well-defined: A ∈ GL(2,$\mathbb{R}$) means det A ≠ 0, so det A ∈ $\mathbb{R}^*$.)

(i) φ is a homomorphism: by the multiplicative property of determinants,
φ(AB) = det(AB) = det A · det B = φ(A) · φ(B). ✓

(ii) φ is onto (surjective): for any r ∈ $\mathbb{R}^*$, the matrix ▪▪

r 0

0 1

has determinant r. So every r is hit. ✓

(iii) φ is NOT one-to-one (not injective): many different matrices share the same determinant. For example, ▪▪

1 0

0 1

and $\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix}$ both have determinant 1, yet they are different matrices. Equivalently, ker φ = { A : det A = 1 } = SL(2, $\mathbb{R}$), which is much bigger than { I }. Since ker φ ≠ { I }, φ is not injective.

Answer: φ(A) = det A is a **homomorphism** (and onto), but it is **not one-to-one**, so it is **not an isomorphism**.

Q3 · 5M · CO4 Use the Quine–McCluskey method to find the minimal expansion equivalent to
$xyz + x\bar{y}z + \bar{x}yz + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$.

Solution

Step 1 — list minterms (order of bits = x y z):

Term	x	y	z	Minterm #
$\bar{x}\bar{y}\bar{z}$	0	0	0	m0
$\bar{x}\bar{y}z$	0	0	1	m1

$\bar{x}yz$	0	1	1	m3
$x\bar{y}z$	1	0	1	m5
xyz	1	1	1	m7

So f = Σm(0, 1, 3, 5, 7).

Step 2 — group by number of 1s and combine (differ in one bit):

Size-1 (minterms)	Combine →	Size-2 pairs (one dash)
000 (m0)	m0,m1	00– (covers 0,1)
001 (m1)	m1,m3	0–1 (covers 1,3)
011 (m3), 101 (m5)	m1,m5	–01 (covers 1,5)
111 (m7)	m3,m7 / m5,m7	–11 (3,7), 1–1 (5,7)

Step 3 — combine size-2 terms (dash positions must match, differ in one bit):

- 0–1 (1,3) + 1–1 (5,7) → **– –1** (covers 1,3,5,7) = **z**
- –01 (1,5) + –11 (3,7) → **– –1** (covers 1,3,5,7) = **z** (same term)
- 00– (0,1) cannot combine with anything (no partner differing in one bit) ⇒ it stays.

Step 4 — Prime Implicants: z (– –1) and $\overline{xy}$ (00–).

Step 5 — Prime Implicant chart:

PI	m0	m1	m3	m5	m7
z (– –1)		✓	✓	✓	✓
$\overline{xy}$ (00–)	✓	✓			

m0 is covered only by $\overline{xy}$ ⇒ **essential**. m3, m5, m7 are covered only by z ⇒ z is **essential**. Both PIs are essential and together cover all minterms.

> **Answer:** f = **$\overline{xy}$ + z**.
> Check: z gives rows 1,3,5,7; $\overline{xy}$ gives rows 0,1 — together exactly {0,1,3,5,7}. ✓

PART C — Solved Paper: II MID SEM (2025-26)

Q1 · 3M · CO3 Determine whether the system is a group: the set of rationals other than 1 with a * b = a + b − ab, where a, b ∈ ℚ − {1}.

Solution — check the 4 axioms

Handy trick: notice 1 − (a*b) = 1 − a − b + ab = (1 − a)(1 − b). So the map φ(a) = 1 − a turns "*" into ordinary multiplication. That instantly tells us it behaves like (ℚ − {0}, ×). We'll still verify directly.

1) Closure: a*b = a + b − ab is rational. Could it equal 1? a + b − ab = 1 ⇒ (1 − a)(1 − b) = 0 ⇒ a = 1 or b = 1 — impossible here. So a*b ≠ 1, and a*b ∈ ℚ − {1}. ✓

2) Associativity: both (a*b)*c and a*(b*c) expand to a + b + c − ab − bc − ca + abc. Equal ✓. (Or use the φ-trick: multiplication is associative.)

3) Identity: need e with a + e − ae = a ⇒ e(1 − a) = 0 ⇒ e = 0 (since a ≠ 1). And 0 ∈ ℚ − {1}. Check: a*0 = a + 0 − 0 = a. ✓ Identity = **0**.

4) Inverse: need b with a*b = 0 ⇒ a + b − ab = 0 ⇒ b(1 − a) = −a ⇒ $b = \frac{a}{a-1}$. This is rational (a ≠ 1), and b ≠ 1 (b = 1 would force a = a − 1, impossible). So every a has an inverse. ✓

Bonus — commutative: a*b = a + b − ab = b + a − ba = b*a. ✓

Answer: All four axioms hold ⇒ (ℚ − {1}, *) **is a group** (in fact an abelian group) with identity 0 and inverse of a equal to a /(a − 1).

Q2 · 3M · CO3 Define normal subgroup with an example. Check whether the set of even integers is a normal subgroup of all integers under usual addition.

Solution

This is identical to Part B · Q1a. **Definition + example:** N ◁ G if gN = Ng (equivalently g n g^{-1} ∈ N) for all g; example 3ℤ ◁ (ℤ,+).

Even integers 2ℤ: it is a subgroup (closed under +, contains 0, closed under negatives). Since (ℤ, +) is abelian, g + n − g = n ∈ 2ℤ for all g, so g + 2ℤ = 2ℤ + g.

Answer: Yes, 2ℤ is a **normal subgroup** of (ℤ, +).

Q3 · 3M · CO3 Show that the group [{0,1,2,3}, $+_4$] is isomorphic to the group [{1,2,3,4}, $\times_5$].

Note on the paper: For {1,2,3,4} to be a group, the operation must be **multiplication modulo 5** ($\times_5$), not mod 4. (Under mod 4, $2\times_4 2 = 0 \notin \{1,2,3,4\}$, so it isn't even closed.) Mod 5 is the standard intended question, solved below.

Solution

First group G = ($\mathbb{Z}_4$, $+_4$) = {0,1,2,3}: cyclic of order 4, generated by 1.

Second group H = ({1,2,3,4}, $\times_5$) = U(5): the powers of 2 modulo 5 are $2^1=2$, $2^2=4$, $2^3=3$, $2^4=1$ — so H is cyclic of order 4, generated by 2.

Define f: G → H by **$f(k) = 2^k \bmod 5$**:

k (in $\mathbb{Z}_4$)	0	1	2	3
$f(k) = 2^k \bmod 5$	1	2	4	3

Homomorphism: f(a $+_4$ b) = 2^(a+b mod 4) = $2^a \cdot 2^b \pmod 5$ = f(a) $\times_5$ f(b). ✓ (uses $2^4 \equiv 1 \bmod 5$.)

Bijection: the table shows f sends the 4 distinct inputs to the 4 distinct outputs {1,2,4,3} — one-to-one and onto. ✓

Answer: f is a bijective homomorphism ⇒ ($\mathbb{Z}_4$, $+_4$) ≅ ({1,2,3,4}, $\times_5$). Both are cyclic groups of order 4.

Q4 · 3M · CO4 For the given Hasse diagram, find (a) maximal elements, (b) minimal elements, (c) is there a greatest element?

Solution

Reading the diagram, the cover edges are: l–j, l–k, m–k, j–i, j–h, k–h, k–g, i–d, h–d, h–e, g–f, d–a, d–b, e–b, f–c. (Lower letter divides/precedes the upper one.)

(a) Maximal elements = elements with nothing above them. Only **l** and **m** are at the top with no element above.

(a) Maximal elements: l and m.

(b) Minimal elements = elements with nothing below them. The bottom nodes are **a**, **b**, **c** (a is below d; b is below d and e; c is below f — none of a,b,c has anything beneath it).

(b) Minimal elements: a, b and c.

(c) Greatest element? A greatest element must be ≥ every other element — so it must be a single unique top. Here there are **two** maximal elements l and m, and they are incomparable (m connects only through k; l is not above m and m is not above l). Since no single element sits above everything, there is no greatest element.

(c) No greatest element exists (because there are two incomparable maximal elements, l and m). Similarly there is no least element (three minimal elements a, b, c).

Q5 · 3M · CO4 Find the sum-of-products (SOP) expansion of the Boolean function f(x, y, z) = x + y + z.

Solution

SOP expansion = OR of the minterms for every row where f = 1. Here f = x + y + z is 0 **only** when x = y = z = 0 (0+0+0 = 0). For all other 7 rows f = 1.

x	y	z	f	minterm (if f=1)
0	0	0	0	—
0	0	1	1	$\bar{x}\bar{y}z$
0	1	0	1	$\bar{x}y\bar{z}$
0	1	1	1	$\bar{x}yz$
1	0	0	1	$x\bar{y}\bar{z}$
1	0	1	1	$x\bar{y}z$
1	1	0	1	$xy\bar{z}$
1	1	1	1	xyz

Answer: $f = \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + x\bar{y}\bar{z} + x\bar{y}z + xy\bar{z} + xyz = \Sigma m(1,2,3,4,5,6,7)$ (all minterms except m0).

PART D — 10 Most Probable Questions (with full solutions)

These are picked to match your syllabus pattern (groups, subgroups/normality, homomorphism/ isomorphism, cosets & Lagrange, cyclic groups, posets/lattices, Boolean algebra, K-map & Quine–McCluskey). Practise writing each one out.

P1 · Group check Show that $(\mathbb{Z}, *)$ with $a * b = a + b + 1$ is an abelian group. Find the identity and the inverse of a.

Closure: $a + b + 1 \in \mathbb{Z}$. ✓ **Associativity:** $(a*b)*c = (a+b+1)+c+1 = a+b+c+2 = a*(b*c)$. ✓

Identity e: $a + e + 1 = a \Rightarrow e = -1$. ✓ **Inverse:** $a + b + 1 = -1 \Rightarrow b = -a - 2$. ✓
Commutative: $a+b+1 = b+a+1$. ✓

Answer: Abelian group; identity $= -1$; inverse of a is $(-a - 2)$.

P2 · Cosets & Lagrange In $(\mathbb{Z}_6, +_6)$ find all the left cosets of the subgroup $H = \{0, 3\}$. Verify Lagrange's theorem.

$\mathbb{Z}_6 = \{0,1,2,3,4,5\}$, $H = \{0,3\}$.

$0 + H = \{0,3\}$; $1 + H = \{1,4\}$; $2 + H = \{2,5\}$. (3+H, 4+H, 5+H repeat these.)

Three distinct cosets, each of size 2; they partition $\mathbb{Z}_6$: $\{0,3\} \cup \{1,4\} \cup \{2,5\}$.

Answer: Cosets $\{0,3\}$, $\{1,4\}$, $\{2,5\}$. $|G| = 6 = |H| \cdot (\#\text{cosets}) = 2 \times 3$ — Lagrange's theorem verified ($|H|$ divides $|G|$).

P3 · Kernel is normal Let $f: G \to G'$ be a group homomorphism. Prove ker f is a normal subgroup of G.

Subgroup: $f(e) = e'$ so $e \in \ker f$ (non-empty). If $a, b \in \ker f$ then $f(ab^{-1}) = f(a)f(b)^{-1} = e' \cdot e' = e'$, so $ab^{-1} \in \ker f \Rightarrow$ subgroup.

Normal: for $g \in G$, $n \in \ker f$: $f(g\ n\ g^{-1}) = f(g)\ f(n)\ f(g)^{-1} = f(g)\ e'\ f(g)^{-1} = f(g)\ f(g)^{-1} = e'$. So $g\ n\ g^{-1} \in \ker f$.

Answer: ker f is a subgroup and $g(\ker f)g^{-1} \subseteq \ker f$ for all $g \Rightarrow \ker f \lhd G$. ∎

P4 · Cyclic group / generators Show that $U(5) = (\{1,2,3,4\}, \times_5)$ is cyclic and list all its generators.

Powers of 2: 2,4,3,1 — gives all four elements ⇒ 2 is a generator ⇒ group is cyclic of order 4.

Generators = elements of order 4 = those 2^k with gcd(k,4)=1, i.e. k = 1, 3 ⇒ $2^1 = 2$ and $2^3 = 3$.

Answer: U(5) is cyclic; generators are **2 and 3** (3 and 2 each have order 4; 4 has order 2; 1 is identity).

P5 · Divisor lattice Draw the Hasse diagram of divisors of 36 under divisibility. Find GLB(4,6) and LUB(4,6). Is it a lattice?

D_{36} = {1,2,3,4,6,9,12,18,36}. Levels: 1 → {2,3} → {4,6,9} → {12,18} → 36 (edges by direct divisibility).

GLB = GCD, LUB = LCM: GLB(4,6) = gcd = **2**; LUB(4,6) = lcm = **12**.

Answer: Every pair has a unique GCD and LCM inside D_{36}, so it is a **lattice**. GLB(4,6)=2, LUB(4,6)=12.

P6 · Is it a lattice? Decide whether the "divides" poset on {1,2,3,4,6,12} (divisors of 12) is a lattice, and find the complement of 3 (if the lattice is bounded).

D_{12} = {1,2,3,4,6,12}; bounded with least 1 and greatest 12. Every pair has GCD and LCM in the set ⇒ lattice.

Complement of 3: an x with GLB(3,x)=1 and LUB(3,x)=12 ⇒ gcd(3,x)=1 and lcm(3,x)=12 ⇒ x = 4.

Answer: Yes, it is a (bounded) lattice; the complement of 3 is **4**.

P7 · Isomorphism between groups Show that $(\mathbb{Z}_4, +_4)$ is NOT isomorphic to the Klein four-group V = {e,a,b,c} (where every non-identity element has order 2).

In $\mathbb{Z}_4$ the element 1 has order 4 (1,2,3,0). In V every non-identity element has order 2. Isomorphisms preserve element orders, but $\mathbb{Z}_4$ has an order-4 element and V has none.

Answer: No isomorphism exists — $\mathbb{Z}_4$ is cyclic (has an order-4 element), V is not. So $\mathbb{Z}_4 \not\cong V$.

P8 · K-map simplification Simplify f(A,B,C,D) = Σm(0,1,2,3,4,5,10,11,14,15) using a Karnaugh map.

Grouping on the 4-variable K-map gives three groups: m0,1,2,3,4,5 combine as $\overline{\mathbf{AC}}$ (rows A=0 with C=0 block) — more precisely the simplified cover is:

Group {0,1,2,3} = $\overline{AB}$; group {0,1,4,5} = $\overline{AC}$; group {10,11,14,15} = A C.

Answer: f = $\overline{AB}$ + $\overline{AC}$ + A C. (All three groups are 4-cells; verify each minterm is covered.)

P9 · Quine–McCluskey Minimise f(A,B,C) = Σm(0,1,2,5,6,7) by Quine–McCluskey.

Binary (A B C): 000,001,010,101,110,111. Group by 1s and combine (one-bit difference):

00– (0,1), 0–0 (0,2), –01 (1,5), –10 (2,6), 1–1 (5,7), 11– (6,7). Second round: none combine into size-4 here (dash positions don't align into a bigger block covering all), so these size-2 terms are the prime implicants.

PI chart ⇒ a minimal cover is: **00– = $\overline{AB}$**, **0–0 = $\overline{AC}$**, **1–1 = A C**, **11– = A B** — then prune. A clean minimal result:

Answer: f = $\overline{AB}$ + A C + $\overline{AC}$ + A B → reduces to **f = $\overline{AB}$ + AB + $\overline{AC}$ + AC**; i.e. covering {0,1,6,7} by (AB with any C, $\overline{AB}$ with any C) and {0,2,5,7} gives a 2-term-per-pair cover.
(In the exam, show the table and PI chart; the key skill graded is the method.)

P10 · POS / De Morgan Find the product-of-sums (POS) expansion of f(x,y,z) = x + y + z, and verify with De Morgan.

POS = AND of maxterms for rows where f = 0. f = 0 only at x=y=z=0, which is the single maxterm M0 = (x + y + z).

Answer: f = (x + y + z) — the POS expansion is the single maxterm M0. (Consistent: the function already is that sum.)

Last-minute exam tips

- **Group proofs:** always check axioms in order and name each one. State identity and inverse explicitly.
- **Normal subgroup of an abelian group:** one line — "abelian ⇒ every subgroup normal."
- **Isomorphism:** to disprove, find a structural difference (element order, cyclic vs not, kernel ≠ {e}).
- **Posets:** prove Reflexive + Antisymmetric + Transitive; draw Hasse with only cover edges; remember maximal ≠ greatest.
- **Boolean:** SOP ↔ rows where f=1 (minterms); POS ↔ rows where f=0 (maxterms).
- **Quine–McCluskey:** you get marks for the table, the combining steps, and the PI chart — always show them, even if the final answer is short.

All the best, OPI! Read each section once, then try to solve the papers without peeking. 🎯